

CATALIN URSU

UNITY GAMEPLAY ENGINEER

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SUMMARY

Unity Gameplay Engineer with 6+ years of experience.

Focused on gameplay systems, architecture, and performance optimization.

Experienced in building scalable systems using Unity, C#, and DI (Zenject), with a strong UI development background.

KEY ACHIEVEMENTS

- Designed scalable architecture using **Zenject (DI)** and async patterns (**UniTask**), improving maintainability and development speed
- Increased framerate from **~40 FPS** to **~70 FPS** by optimizing CPU/GPU load and eliminating bottlenecks (**Unity Profiler, Frame Debugger, Memory Profiler**)
- Reduced memory usage by **30% (1.08GB → 0.88GB)** using Addressables, asset compression, and optimized resource management
- Built resolution-independent UI system with Safe Area support, ensuring compatibility across mobile devices (4:3–21:9)
- Improved application responsiveness by redesigning async workflow (**UniTask, R3**), eliminating frame delays and unnecessary allocations

SKILLS

Core: Unity, C#, Gameplay Systems, UI Systems

Architecture: SOLID, DRY, Zenject (Dependency Injection), Event-driven architecture

Async & Reactive: UniTask, R3 (ReactiveProperty)

Performance: Unity Profiler, Frame Debugger, Memory Profiler, CPU/GPU Optimization

Assets: Addressables, Asset Optimization (Textures, Audio), Resource Management

UI: UGUI, Adaptive Layouts, Safe Area, Render Textures

Additional: Multiplayer (Netcode for GameObjects), Mobile Development, VR/AR

WORK EXPERIENCE

Bully Entertainment — Unity Gameplay Engineer

Apr 2021 – PRESENT

- Led architecture design and implemented modular systems using DI and async patterns
- Developed gameplay systems and complex UI architectures for production projects
- Reworked and adapted complex UI systems for new designs and additional content, improving maintainability and extensibility
- Optimized performance (**CPU/GPU, batching, memory**)
- Built reusable UI systems and components, reducing development time for new features
- Implemented async-driven UI logic using **R3**, eliminating unnecessary Update loops and reducing memory allocations
- Designed and implemented DLC system with card-based mechanics, including deck generation logic with guaranteed win conditions
- Developed reusable subtitle system (**DoTween, TextMeshPro**) with dynamic timing based on audio playback
- Implemented multiplayer synchronization logic for **cross-platform MR** project (VR + PC), ensuring consistent shared world state
- Collaborated in a team of 6–7 members (2 developers), contributing to architecture decisions and code quality

Midnight Works — Unity Developer

Jun 2020 – Sep 2020

- Developed gameplay prototypes for 2D and 3D projects
- Implemented core gameplay mechanics and UI systems
- Integrated monetization systems (AdMob, Facebook SDK)
- Optimized performance for mobile devices

PROJECTS

Showcase - Gameplay Pet Project (<https://github.com/CatalinUrsu/Showcase>)

- Developed a complete 2D game focused on gameplay systems and architecture
- Designed scalable architecture using Zenject (DI), UniTask, and R3
- Implemented reactive UI interaction logic using event-driven approach
- Focused on clean code, extensibility, and performance optimization

UI Systems & Interaction Logic

- Rebuilt UI interaction logic using event-driven and reactive patterns
- Eliminated Update-based logic and unnecessary allocations
- Implemented snapping, inertia, and optimized scrolling behavior

Performance Optimization

- Conducted deep profiling and optimization using Unity's Profiling Tools
- Focused on performance and efficient system design
- Improved runtime stability and performance

Tools & Systems (Pet Projects) ([Project - Helpers](#), [Project - IdleNumber](#))

- Developed reusable helper library for Unity projects
- Implemented large number system for idle games
- Focused on reusability, performance, and clean architecture

EDUCATION

State University of Moldova (2017-2020)

Computer Science

LANGUAGES

Romanian — Native

Russian — Native

English — B2 (Upper Intermediate)